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Coiled Tubing Intervention in Extreme Sour Wells (H_2S 40%) to Prevent Sulfide Stress Cracking: Best Practices

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Abstract

The need to maximize hydrocarbon extraction from complex reservoirs is a key driver for the coiled tubing (CT) market, which offers efficient rigless well intervention solutions. A significant challenge lies in performing acid stimulation in severely sour wells with high temperatures, while simultaneously protecting the coiled tubing from corrosion and sulphide stress cracking (SSC). This study presents guidelines for the successful execution of CT interventions in extreme conditions containing up to 40% H_2S and 11% CO_2, under reservoir pressures of 6,000 psi and bottomhole temperatures near 300°F, all while mitigating risks associated with aci d and H_2S corrosion.

To address these conditions, an integrated approach was developed combining tailored pipe integrity management, recommended CT practices in sour services (API RP 5C7) and chemical protection strategies. This method establishes a comprehensive framework for operating CT in aggressive sour environments while extending its service life. The pipe management system continuously updates the CT's operational limits, monitors wall thickness, and characterizes deformations. It also evaluates pipe condition by detecting anomalies—such as pitting, mechanical damage, and corrosion—in accordance with API RP 5C8. By analysing extensive integrity data in real time, the system supports proactive decision-making, tracks the evolution of defects, and dynamically adjusts operating parameters.

Chemical protection, incorporating specialized scavengers and fatigue derating, is used to inhibit corrosion in conjunction with engineered pipe management. This paper highlights how improved practices and continuous monitoring of defect progression throughout the string's lifespan enhance reliability and allow for on-the-fly adjustments to the operating window, reducing the risk of tubing failure. Case studies are included to illustrate key practices and provide lessons for the design and execution of CT interventions in similarly demanding conditions.

Introduction

Many prolific hydrocarbon reservoirs contain significant concentrations of H_2S, posing severe risks of toxicity, corrosion, and embrittlement (Kermani & Harrop, 1996). Combining CT operations with acidic fluids in an H_2S environment creates a highly aggressive scenario promoting multiple failure mechanisms:

Hydrogen-Induced Cracking (HIC), and Sulfide Stress Cracking (SSC) (Elboujdaini & Revie, 2009). The evolution of CT for acid stimulation in sour H_2S wells is a compelling narrative of overcoming extreme technical challenges through relentless innovation. Driven by catastrophic early failures, the industry embarked on a path defined by profound advancements in metallurgical science, leading to the development and qualification of sophisticated low-alloy steels and ultimately Corrosion Resistant Alloys (CRAs) capable of withstanding the combined assault of acid and H_2S. Parallel developments in acid chemistry, particularly high-performance corrosion inhibitors and H_2S scavengers, and the establishment of rigorous standards (NACE/ISO) and safety protocols, formed the bedrock for safe operations. Early CT was predominantly low-alloy carbon steel (e.g. QT-700, QT-800). Field experience quickly revealed catastrophic failures in sour service due to SSC and HIC (Stanley & Gentili, 1986). Acid exposure exacerbated these issues, dissolving protective scales and exposing fresh metal to H_2S attack. Research solidified understanding of SSC (brittle failure under tensile stress in H_2S) and HIC (stepwise cracking along inclusions due to absorbed hydrogen). NACE standards like TM0177 (Laboratory Testing) and MR0175 (Material Requirements) began defining environmental severity (Partial Pressure H_2S, pH) and acceptable material performance limits (Fig. 1). Initially, derating CT yield strength (using QT-700 instead of QT-1000) was common practice to reduce SSC susceptibility, sacrificing collapse and tensile capacity (Tipton, 1994). Development of filming amine-based corrosion inhibitors intensified, though effectiveness in dynamic CT conditions with H_2S and acid was often inconsistent and required high concentrations (Frenier, 2001). Strict limitations on H_2S partial pressure, fluid pH control, and reduced operating pressures/tensions were imposed, significantly restricting CT's applicability in demanding sour wells (Patton, 1990). The development of major sour fields (e.g., Kashagan, Tengiz, deep HPHT Gulf of Mexico) demanded CT capable of handling high pressures, temperatures, and severe H_2S. The improved manufacturing (cleaner steel, controlled rolling, optimized quenching and tempering) produced grades like QT-900, QT-1000, and proprietary equivalents (e.g., "Sour Service Grade") meeting NACE MR0175/ISO 15156 requirements for specific environmental conditions (H_2S pp, pH, temperature) (Craig, 2003; PRCI, 2005). These balanced higher strength with SSC/HIC resistance.

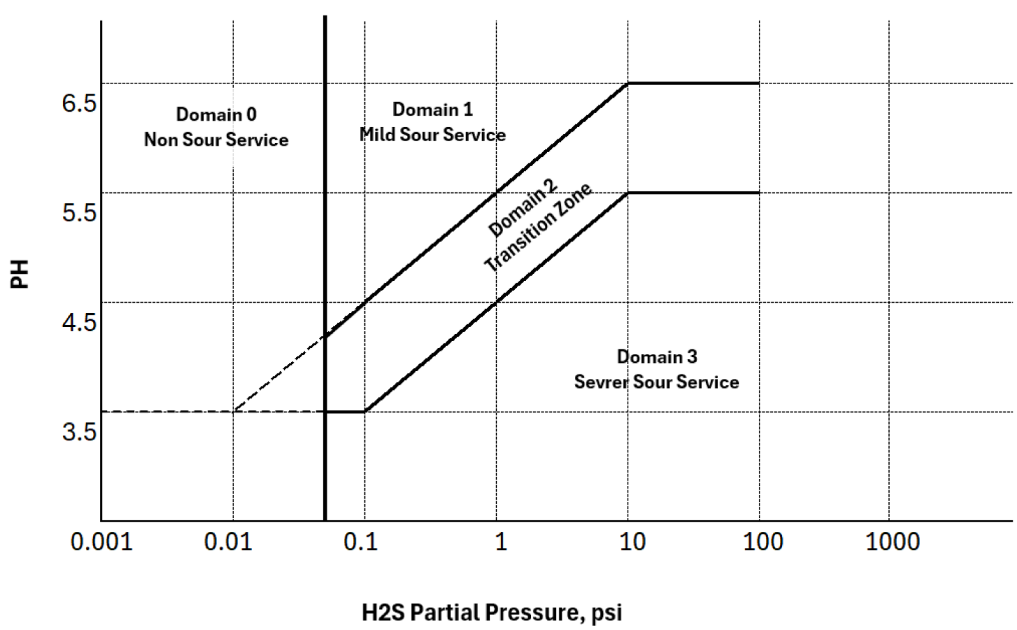

Figure 1—Sour Service Domains by NACE MR0175/ISO 15156

The application of CT for acid stimulation treatment in wells containing hydrogen sulfide (H_2S) - sour wells - represents a nexus of technological advancement driven by harsh operational demands. Tracing developments from early material failures to contemporary high-integrity systems, the review highlights key milestones:

- The critical understanding of sulfide stress cracking (SSC) and its mitigation through advanced alloys (Kermani & Morshed, 2003, Tiratsoo, 2010 and Agha et al., 2012).
- The integration of effective, compatible H_2S scavengers (aldehyde-based, metal carboxylates) into acid blends or as pre-/post- flushes under simulated downhole conditions (T, P, H_2S, flow regime) became critical to reduce dissolved H_2S concentration during treatment (Jovancicevic et al., 1999, Nasr-El-Din & Samuel, 2007 and Kelland, 2009).
- Enhanced Operational Practices and Safety Protocol, detailed risk assessments (HAZID, HAZOP) specific to H_2S and CT acidizing became standard, covering material selection, fluid compatibility, contingency plans, and emergency response (API RP 5C7, 2017); and nitrogen lifting during pre-flush or post-flush to evacuate spent acid and H_2S from the CT string to minimize exposure time (Craig & Brown, 2018). NACE MR0175/ISO 15156 became the cornerstone for material selection. Rigorous qualification testing (SSC, HIC per NACE TM0177 and TM0284 methods) became mandatory for sour service CT (ISO 151562, 2015). API RP 5C7 (Coiled Tubing Operations) incorporated sour service guidelines; and
- The refinement of H_2S safety protocols and real-time monitoring analysis enabled dynamic adjustments, has provided unprecedented control and optimization capabilities during the job, maximizing efficiency and safety in complex sour environments (Al-Arnaout et al., 2019).

The analysis reveals a trajectory defined by relentless innovation in metallurgy, rigorous standards (notably NACE MR0175/ISO 15156), and enhanced operational practices, transforming CT from a risky proposition to a preferred method for controlled acidizing in the world's most challenging sour reservoirs. This paper examines CT technology, materials, operational practices and procedures; and monitoring technologies and acid systems specifically tailored for safe and effective interventions in sour environments. Case histories summarize the guidelines of practices to synthesize findings for establishing the best protection and yield lessons for design and execution of CT intervention for similar unforgiving environments.

Equipment Selection

Corrosion Mechanisms and Environmental Impact

Hampson (2009) emphasized that brines in H_2S or CO_2 environments exacerbate corrosion risks due to the formation of sulfuric acid from H_2S and carbonic acid from CO_2 when dissolved in water. Both acids attack CT pipes, causing corrosion that can compromise operational integrity. Gupta (2005) further elaborated on CO_2-related challenges, stating that while pure, dry CO_2 is non-corrosive, its interaction with water and contaminants like hydrogen sulfide and oxygen create highly corrosive conditions. When water saturated with CO_2 is present, the pH can drop below 4, accelerating corrosion.

Coiled Tubing Grade

The selection of coiled tubing grades for deployment in sour service environments, characterized by the presence of hydrogen sulfide (H_2S), represents a critical engineering challenge demanding meticulous attention to material science, operational me chanics, and stringent industry standards. This selection process must reconcile the inherent mechanical demands of CT operations (cyclic plastic deformation during spooling, high-pressure containment, abrasive wear during wellbore intervention) with the aggressive degradation mechanisms activated by H_2S. The cornerstone standard governing material suitability is NACE

MR0175/ISO 15156 (Materials for use in H_2S-containing environments in oil and gas production), which defines environmental severity limits based primarily on H_2S partial pressure and pH (Fig. 1).

The NACE MR0175/ISO 15156 standard imposes strict hardness limits (typically ≤ 22 HRC, **Rockwell Hardness C**) and often yields strength limits (frequently ≤ 90,000psi or 90 kpsi). Higher grades are generally prohibited for sour service due to th eir high inherent hardness, making them extremely SSC-prone. Table-1, grades CT-90 and below, meets the recommended hardness. McCoy and Thomas (2006) demonstrate that sour service drastically reduces CT fatigue life, specifically, fatigue life can be reduced by up to 50%, which forces operators to limit the usable strength range of CT, despite the inherent advantages of higher-strength grades for deep and high-pressure operations. The reduction was more severe in higher-strength grades (e.g., 110 kpsi vs. 90 kpsi).

Table 1—Yield strength and hardness of common coiled tubing grades

<table>
<tr><th>CT grades</th><th>Yield Strength (psi)</th><th>Hardness (HRC)</th><th>Sour Service Suitability</th></tr>
<tr><td>CT 55</td><td>55,000</td><td>Less than 22</td><td>low risk</td></tr>
<tr><td>CT 70</td><td>70,000</td><td>22</td><td rowspan="2">Acceptable</td></tr>
<tr><td>CT 80</td><td>80,000</td><td>22</td></tr>
<tr><td>CT 90</td><td>90,000</td><td>22</td><td>Marginal</td></tr>
<tr><td>CT 100</td><td>100,000</td><td>28</td><td rowspan="3">Not Recommended</td></tr>
<tr><td>CT 110</td><td>110,000</td><td>30</td></tr>
<tr><td>CT 130</td><td>130,000</td><td>39</td></tr>
</table>

Surface imperfections on CT, including minor marks from handling equipment, can locally increase hardness and raise the risk of Sulfide Stress Cracking (SSC). Therefore, selecting a string with lower accumulated fatigue is preferable to one with extensive use history. Even CT grades typically more resistant to H_2S, like grade 80, have suffered SSC failures when surface flaws or damage exist. Consequently, for critical operations or when CT will face high pressures, operators should consider inspecting the entire string using Non-Destructive Evaluation (NDE). The chosen NDE method must be capable of detecting not just changes in diameter, ovality, and wall thickness, but crucially also external damage and cracks.

CT Bottom Hole Assembly (BHA)

Most connectors used to attach downhole tools to the CT end cause mechanical hardening damage at the connection point (fig. 2), increasing the CT's vulnerability in that area and the susceptibility of the CT to SSC (Hampson, 2009). It's recommended that sections damaged by these connectors be removed between operational runs, especially after exposure to sour wells and make up a new connector for every run. Using manual butt welds in wells with potential H_2S issues is generally discouraged. Concerns include the potential for higher average hardness in the weld's Heat-Affected Zone and inconsistent quality inherent to manual welding processes.

Figure 2—Induced mechanical hardening damage at the connection point

AISI 4140 is a chromium-molybdenum (Cr-Mo) alloy steel known as a "through-hardening" steel. It is widely favored for downhole tools due to its excellent combination of properties:

- High Strength: It can be heat-treated to achieve very high strength and hardness levels, making it suitable for tools that must withstand significant mechanical loads, shock, and abrasion.
- Good Toughness: It maintains good impact resistance and toughness at its high strength levels, which is crucial for tools that may experience sudden loads or shocks downhole.
- Excellent Wear Resistance: The hardness achieved through heat treatment provides excellent resistance to wear from contact with casing, open hole, and debris.
- Machinability: In its annealed (softer) state, 4140 is relatively easy to machine into complex shapes, which is ideal for manufacturing intricate tool components.
- Cost-Effectiveness: It offers a favourable balance of performance and cost compared to more exotic corrosion-resistant alloys.

The limitation of 4140-grade is typically heat-treated to strengths far exceeding this limit. A standard heat-treated 4140 component will have a hardness in the range of 28-32 HRC or higher, making it extremely susceptible to Sulfide Stress Cracking (SSC) in the presence of H_2S. While this material is cost-effective and widely used, advanced materials such as Monel and Inconel are considered for high- H2S environments and acid service due to their superior corrosion resistance.

H_2S Inhibitor Injection

The location for injecting corrosion inhibitor depends on the specific operation and well conditions. There are two primary injection points:

- Injection Sub: Installed within the surface Pressure Control Equipment stack.
- Tee Sub: Used for injecting into the circulating fluid path (either down the coiled tubing itself or down the annulus).

Injection Sub. The purpose of the injection sub is to pump Inhibitor around the outer diameter (OD) of the coiled tubing while running into the well, creating a sufficient coating barrier on the coiled tubing's outer diameter (OD) to separate H_2S from the steel surface for effective protection.

Tee Sub. The injection tee for injecting the inhibitor into the circulating fluid, positioned downstream the nitrogen and/or fluid pumps. This line must incorporate a check valve to prevent backflow of inhibitor towards these pumps. Injecting inhibitor at either the tee location or an injection sub necessitates the use of a pump. Positive-displacement chemical-injection pumps, such as air-powered diaphragm pumps, are typically deployed for this application.

Chemical Inhibitors for Sour Service

Corrosion was evaluated using metal weight-loss measurements on metal samples of an intact coil. Fig. 3 represents the flowing environment of pH and H_2S partial pressure. Testing occurred at 300°F and 1,000 psi pressure, simulating Well's flowing conditions. These samples were exposed to acid solutions replicating field conditions (15% HCL and 20% HCL). Experiments included environments under gas mixtures containing H_2S and CO_2, and trials with a 1 % H_2S scavenger added to the chemical recipes. Exposure to the acidic gases CO_2 and H_2S produced unacceptable corrosion rates of 0.1248 lbm/ft^2. Adding the H_2S scavenger significantly reduced the corrosion rate and metal loss. Results showed acceptable corrosion rates of 0.0403 lbm/ft^2, below the threshold acceptable limit of 0.05 lbm/ft2 (pounds per square feet) in some conditions. These findings demonstrate a highly corrosive environment and highlight the necessity for applying the correct concentration of mitigation chemicals.

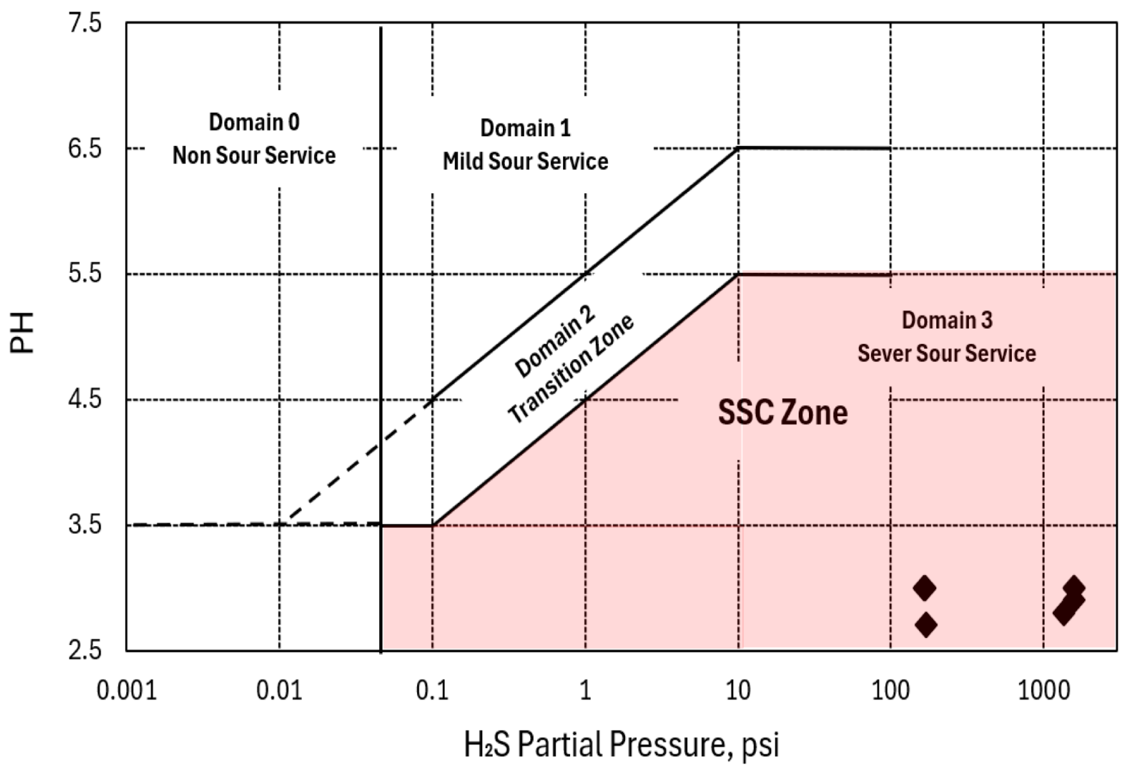

Figure 3—SSC Zone for CT without inhibitor

Minimizing direct exposure of CT to active acids is critically important, particularly under high-temperature conditions with elevated concentrations of acidic gases like H_2S and CO_2. Therefore, pumping H_2S scavengers ahead of the acid stage is crucial. Rigorous assessment of acid-induced corrosion in both spent and active acids must be performed, replicating downhole conditions including pH, H_2S, and CO_2 levels. Contact between the tubing and corrosive fluids should be prevented. An appropriate H_2S scavenger should be introduced into the acid immediately prior to pumping, with its concentration optimized to avoid significant impairment of the acid's reactivity with calcium carbonate (Nasr-El-Din et al., 2000).

Laboratory tests and operational experience demonstrate that chemical inhibitors play a vital role in preventing unexpected failures by mitigating hydrogen uptake and environmental cracking risks in sour environments. Two primary inhibitor types are used:

- Non-triazine-based formulation, combined with corrosion inhibitors in a hydrocarbon solvent. It reacts quickly and efficiently with dissolved H_2S in the fluid stream as it is pumped down the CT and returning to the annulus. This rapidly converts toxic, corrosive H2S into non-corrosive, soluble by-products before it can significantly attack the CT string or reach surface at dangerous concentrations. It is specifically formulated to work effectively within the aggressive chemical environment of acid stimulation without detrimental interactions or impact acid performance or generate undesirable precipitates.
- Formaldehyde-based blend functioning as both a sulfide scavenger and a cracking inhibitor specifically for acid treatments. It reacts with H_2S (scavenging) to form a protective coating on the pipe. For optimal effectiveness, it should be added to the acid blend immediately before pumping ("on-the-fly"). Formaldehyde (HCHO) reacts instantly with H_2S to form water-soluble trithiane:

$$3HCHO + H_2S \rightarrow (CH_2)_3S_3 + H_2O$$

This rapid reaction minimizes CT exposure to corrosive H_2S during pumping/flowback.

Operational Procedures

During Run-in-Hole (RIH), continue pumping inhibitor to coat the outer OD of the CT via the annulus while RIH, calculate and set the displacement pump rate, using pump stroke/displacement as a guide to ensure stable injection.

At target depth, pump the appropriate laboratory tested inhibitor within the treatment fluid at the specified concentration. This provides additional protection as inhibitor-containing fluid/gas circulates-up the annulus.

With Nitrogen, when pumping nitrogen lifting alone, periodically inject inhibitor into the nitrogen stream in batches of 5 bbls every 3 hours.

Pressure Control Equipment

The selection and configuration of Pressure Control Equipment (PCE) – the critical barrier between uncontrolled reservoir energy and the surface environment – demands an uncompromising, standards-driven engineering approach far exceeding requirements for sweet service. Failure modes include Sulfide Stress Cracking (SSC), Hydrogen Embrittlement (HE), accelerated corrosion, and elastomer degradation, any of which can result in loss of containment, uncontrolled release of H_2S, potential fatalities, sig nificant environmental damage, and asset loss. This section details the critical technical considerations for PCE selection and configuration to mitigate these severe risks (Table-2).

Table 2—PCE Configuration Philosophy for Sour Wells

Material Compliance	All wet components must comply with NACE MR0175/ISO 15156. This dictates material grades, hardness limits, and heat treatments.
Elastomer Compatibility	Seals and elastomers must be specifically rated for H_2S service and compatible with wellbore fluids (oil, gas, brine, acids). Perfluoro elastomer (FFKM) offer the highest chemical resistance to aggressive environments, offers excellent chemical resistance to H_2S
Enhanced Sealing Integrity	The API Standard 53 (2015) recommends that the connection of the blowout preventer (BOP) stack should be flanged with metal-to-metal seals. Connections: API specification 6A flanged connection: 10K or 15K with NACE ring grooves/gaskets. Rated ≥ MASP.
Corrosion Management	Consider potential for localized corrosion (pitting, SSC) under H_2S/CO_2 conditions. Material selection and potential inhibition plans are vital.
Function Testing	Equipment must be function-tested rigorously before entering the sour environment, often under simulated conditions if possible. Control System Remote operator console positioned in safe zone (upwind). ESD: Multiple, clearly marked ESD buttons activating all BOP functions instantly.
Documentation & Certification	Meticulous traceability of material certifications (MTRs) for all wet components and third-party inspection reports are essential. All components must have material certifications proving compliance with NACE MR0175/ISO 15156.

Engineering Requirements and Configuration

The API Standard 53 (2015) recommends that the lowermost connection of the blowout preventer (BOP) stack should be flanged as a minimum standard. However, considering the elevated H_2S and CO_2 concentrations in modern projects, the engineering team expanded this requirement. The updated protocol includes flanged connections throughout the entire PCE, extending from the bottom connection of the lowermost BOP to the stripper. Additionally, a backup BOP and stripper are integrated into the stack to enhance safety and redundancy for these critical interventions (Fig. 4).

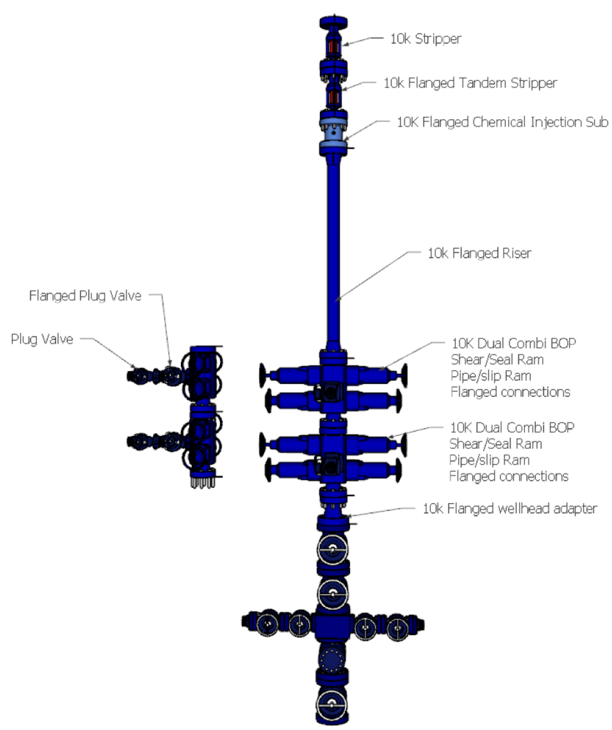

Figure 4—Pressure control equipment (PCE) configuration

Material Selection: The Foundation of Integrity

Compliance with NACE MR0175/ISO 15156 (Materials for use in H^S-containing environments in oil and gas production) is not merely recommended; it is mandatory for all pressure-containing and critical load-bearing components. Material suitability depends on:

- Partial Pressure of H_2S: The primary driver for material class selection.
- In-situ pH: Lower pH increases susceptibility to SSC.
- Chloride Content: Higher chlorides exacerbate pitting/crevice corrosion, impacting SSC resistance.
- Temperature: SSC susceptibility is highest in the "critical" range (typically ambient to ~80°C / 175°F)
- Hardness: A critical parameter strictly controlled by NACE; harder materials are more susceptible.

Non-Metallic Components (Elastomers and Seals)

Coiled Tubing (CT) pressure control equipment (PCE) — including blowout preventers (BOPs), strippers, and connectors—must maintain integrity in sour wells under extreme conditions (300°F, 5,000 psi). Seal failure risks catastrophic leaks, H_2S exposure, and environmental incidents. This section outlines a rigorous selection methodology aligned with NACE MR0175/ISO 15156-3, API 16ST, and field-proven practices. Perfluoroelastomer (FFKM) offer the highest chemical resistance to aggressive environments, offers excellent chemical resistance to H_2S. Hydrogenated Nitrile Butadiene Rubber (HNBR) offers good oil resistance but generally inferior H_2S resistance compared to premium FFKMs. Avoid standard Nitrile (NBR).

PCE Configuration Philosophy for Sour Wells

The configuration must prioritize redundancy, leak minimization, and rapid isolation. The "Barrier Philosophy" is paramount.

Primary Containment and Control.

- CT Connector: The critical first barrier. Must be rated for well shut-in pressure (WSIP) sealing the CT pipe from wellbore fluids under sour conditions.
- Pipe Rams: Positioned below the shear/seal BOP. Provide a static seal on the CT body. Must be SSC-resistant material.
- Shear/Seal Blowout Preventer (BOP): The single most critical component and positioned just above the flow cross. Must reliably shear the specific CT grade and size and seal the wellbore immediately after shearing under maximum anticipated surface pressure (MASP). Seal the open bore if CT is removed or sheared. Must be SSC-resistant. Requires rigorous function testing and material certification (body, rams, seals). Material compliance is non-negotiable.
- Stripper: Provides dynamic seal on the CT. Requires H_2S-resistant packing elements and body material. Dual stripper configurations are highly recommended for sour wells. Primary stripper handles normal operating pressure. H_2S-rated element. Backup Stripper: Engaged if primary fails or during higher pressure operations (e.g., well control). Also, H_2S rated. Allows safe change-out of primary element under pressure.

Secondary Containment and Flow Path.

- Flow Cross/Manifold: Provides controlled flow paths to choke manifold and kill line. All valves (manual and hydraulic), fittings, and the cross body must be NACE compliant.

- Check Valves: Essential in flow lines to prevent backflow. Use dual, staggered checks rated for sour service. Material selection is critical for internals.
- Surface Safety Valve (SSV): Provides an emergency barrier independent of the CT. Must be NACE compliant.

Tertiary Measures and Safety Systems.

- Leak Detection: Comprehensive system with strategically placed H_2S sensors around the stack, flow lines, and work area. Audible/visual alarms integrated with ESD.
- Emergency Shutdown (ESD) System: Automatically closes SSV, annular (if present), and potentially flowline valves upon H_2S detection, high pressure, or manual activation. Must function reliably in the environment.
- Ventilation: Forced air ventilation (positive pressure in operator areas, extraction near potential leak points) is mandatory.

Operational & Design Considerations.

- Minimize Connections: Every connection is a potential leak point. Use flanged with all metal-to-metal seals connections with proper NACE gaskets over threaded where possible. The use of all-metal-to-metal seals in the stack, replacing quick unions such as Bowen.
- Pressure Testing: Function test BOPs daily at low pressure. Pressure test the entire assembled stack (including flow cross and manifolds) to the rated working pressure or WSIP (whichever is higher) before operations and after any significant change. Use compatible test fluid.
- Corrosion Inhibitor: Continuous injection compatible with elastomers may be required downstream of the stripper.
- Personnel Training & PPE: Rigorous training on H_2S hazards, ESD procedures, and SCBA use. Full H_2S-rated PPE (SCBA on standby) is mandatory for personnel near the stack during operations. Personnel competency is paramount in mitigating risks during coiled tubing interventions. Proper training and adherence to best practices are essential to ensure the safety and success of these operations.
- Procedures: Detailed Job Safety Analysis (JSA) and Hazard Identification (HAZID) specific to the well's sour parameters (H_2S concentration, pressure, temperature). Review Material Certification for all PCE components (Mill Certs, NACE compliance statements). Verify compatibility of all elastomers.
- Blowout Risks and Emergency Preparedness: As noted by Nasr-El-Din and Metcalf (2004), operations in sour wells pose a significant blowout risk, with documented incidents such as those in Canada prompting stricter regulations in populated areas. Additionally, displaced control fluids and free hydrocarbons can create ground fire hazards.
- Using elastomers specifically designed for aggressive environments and resistance to explosive decompression is mandatory for both surface equipment and downhole tools.

To address these risks, blowout contingency plans and emergency response training for personnel are critical. Unlike conventional gas- or oil-well blowouts, blowouts require tailored strategies to account for specific hazards such as phase transitions and the rapid expansion of CO_2. Equipping teams with the knowledge and skills to respond effectively ensures operational safety and minimizes environmental impact. Hydraulic Control System: Accumulator bottles must provide sufficient volume to close all critical functions (SSV, annular, shear rams) with pressure margin. Control fluid compatibility with sour service elastomers must be confirmed.

Real-Time Coiled Tubing Integrity Management

Mechanical defects like cracks, corrosion, wall thinning, and ovality accumulate due to extreme cyclic stresses, chemical exposure, and corrosive downhole environments. Traditional inspection methods often occur offline, requiring downtime, offering delayed results, and potentially missing critical micro-defects. Real time coiled tubing integrity monitor enables providing a continuous inspection during well intervention operation rather than traditional models to update fatigue life and identify CT pipe defects, enabling proactive pipe management to mitigate failures. In field implementation, the real time coiled tubing integrity monitor provided inspection results from up-hole end to downhole end. The analysis aims to determine whether the CT pipe can be used for further operations.

Fig. 6 and 7 show the increase in the outside diameter and the ovality for the specific CT pipe section during high-pressure cycling while performing wiper trips during well intervention operation. Many OD/Ovality alarms detected because of the vibration on the pipe during RIH and POOH. However, The OD was checked during the inspection, and it was matching with the nominal value. The pipe wall thickness monitoring shows no abnormal changes in wall thickness observed along the pipe (Fig.8).

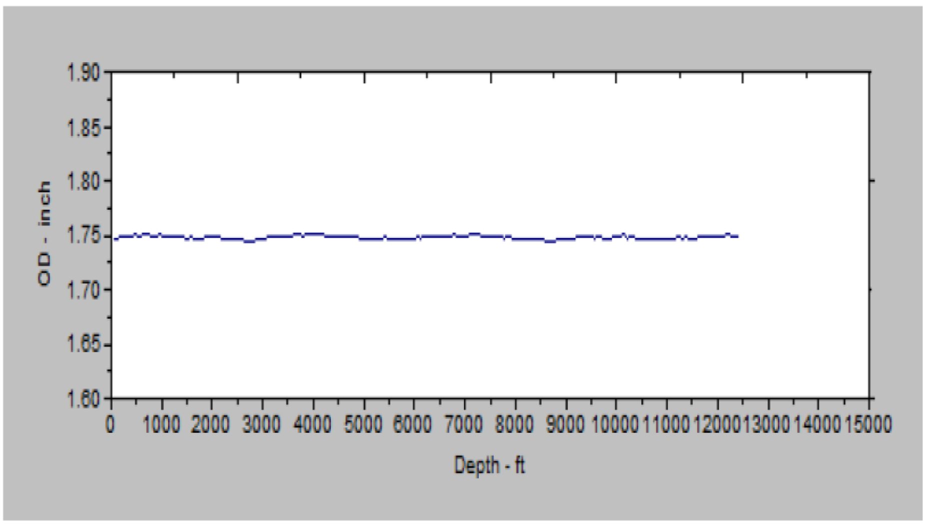

Figure 6—CT string outside diameter measurements.

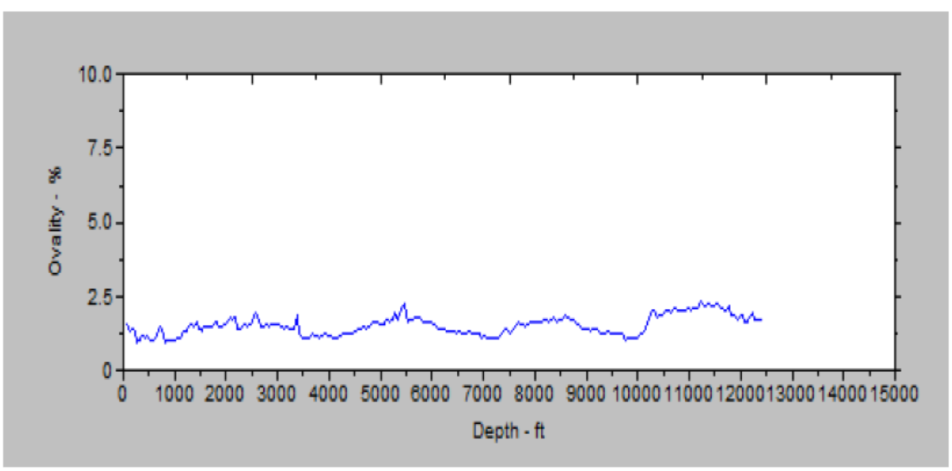

Figure 7—CT string ovality measurements

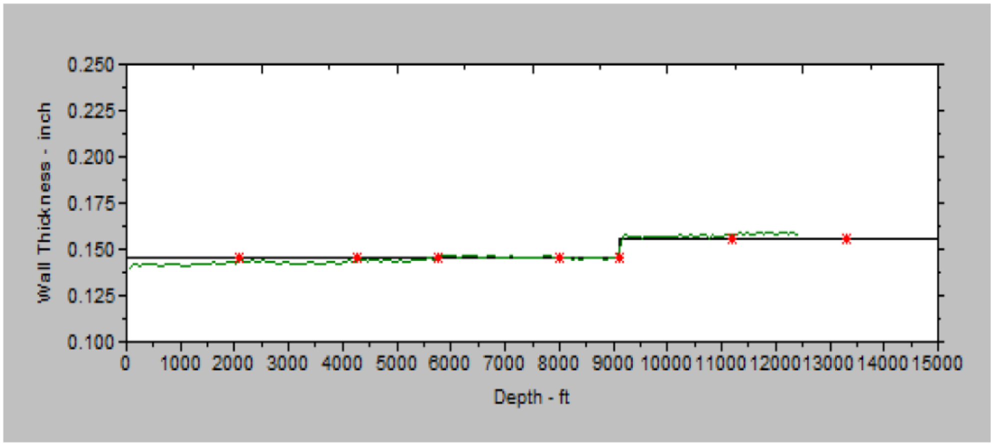

Figure 8—CT wall thickness measurements

Real time CT integrity management monitor (Fig. 9) employs a sophisticated hybrid sensor array deployed directly on the CT string during operations. Its core strength lies in seamlessly integrating multiple Non-Destructive Testing (NDT) principles:

- Advanced Magnetic Flux Leakage (MFL): Detects surface and near-surface anomalies like cracks, pitting corrosion, mechanical gouges, and inclusions with high sensitivity. Refined algorithms distinguish between defect types and suppress noise.
- Ultrasonic Testing (UT): Precisely measures wall thickness (detecting internal/external corrosion and erosion) and ovality (identifying cross-sectional deformation critical for fatigue life assessment). Real-time UT evaluation provides absolute wall thickness data.
- High-Resolution Eddy Current: Enhances detection of surface-breaking flaws and provides accurate diameter measurements.

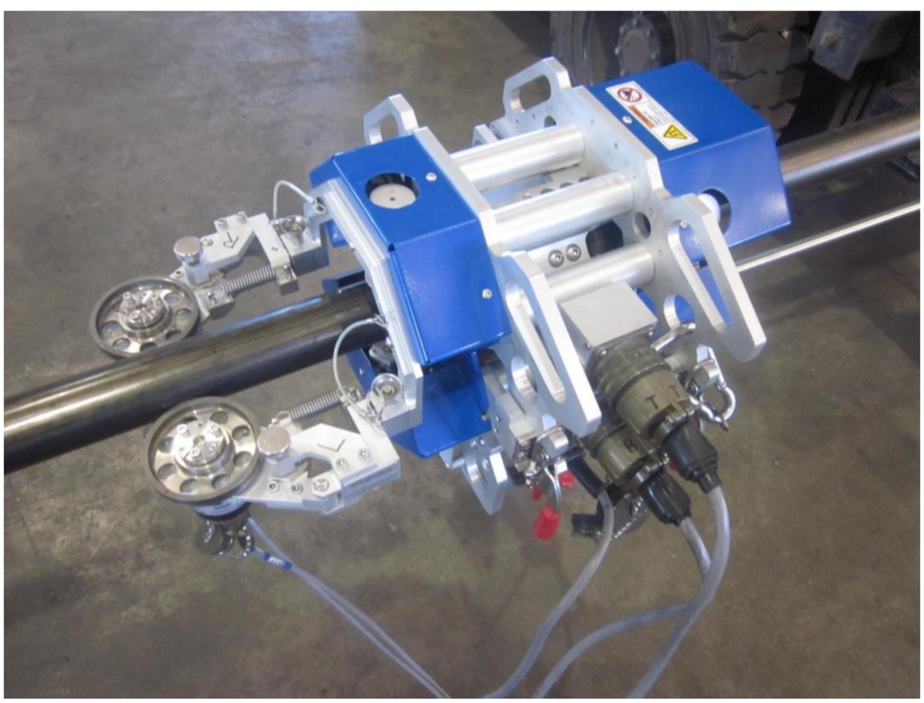

Figure 9—Real Time Coiled Tubing Integrity Monitor

The real time CT inspection analysis was utilized to predict the number of CT cycles under pressure for each section (Fig. 10). Based on the fatigue analysis, 100 cycles remain for the pipe to fail at depth of ~9160ft under 5000psi. However, other cycles are generated based on false alarms which are not counted as a real case. This supports proper utilization of the CT string for further runs and optimizes intervention designs.

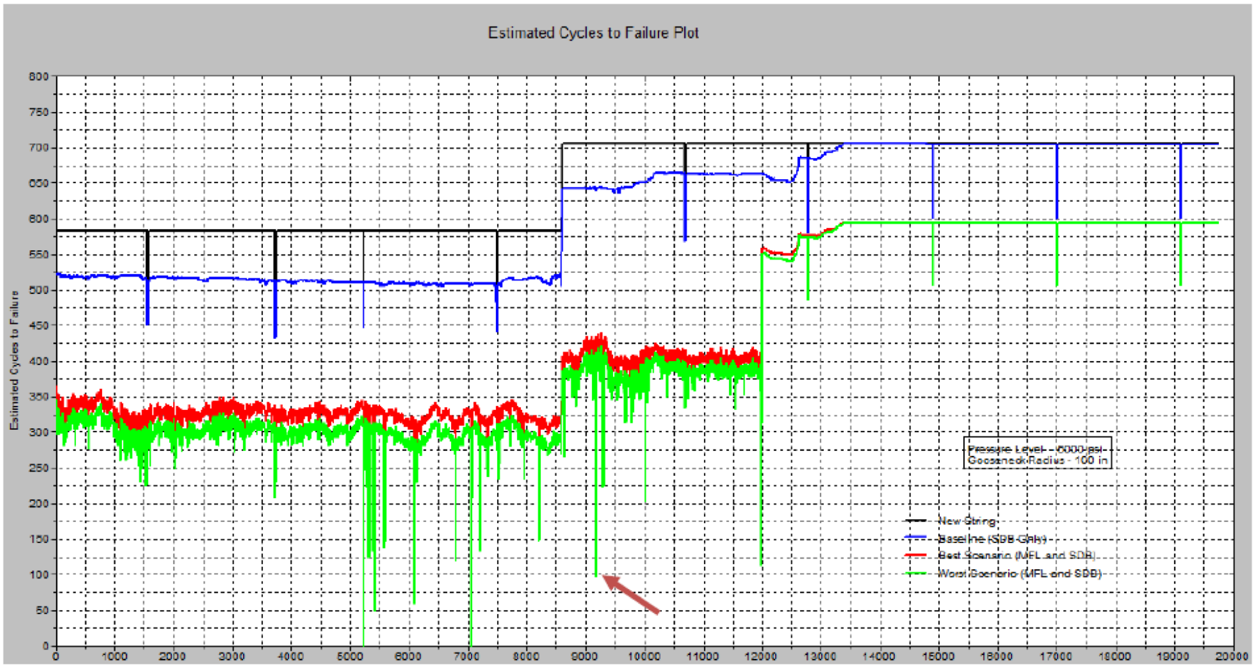

Figure 10—Estimated CT cycles to failure plot.

Real-time visualization empowers field supervisors and engineers to make informed decisions instantly: continue operations safely, adjust parameters, or pull the CT based on actual, up-to-the-minute integrity data. Another benefit is that the historical trending aids in predictive maintenance, facilitating the adherence to inspection requirements outlined in standards like API RP 5C7 and API SPEC 5ST.

Conclusions

Effective risk management during well interventions in high H_2S and CO_2 environments relies on comprehensive personal trainin g, ongoing monitoring, and operational readiness. Upholding strict competency requirements and preventive safety protocols allows operators to control potential hazards, safeguarding both personnel and operational integrity. The use of suitable coiled tubing (CT) materials, along with strong protective and monitoring strategies, is essential for successful interventions in severely sour conditions. Through proactive corrosion control and compliance with industry standards, operators can reduce threats from H_2S and CO_2, maintaining operational safety and efficiency even with standard CT grades.

The choice, setup, and safeguarding of pressure control equipment under extreme sour conditions require compliance with advanced engineering standards and specialized materials. By integrating strong safety measures and standard operating procedures (SOPs), operators can lower risks related to H_2S and CO_2 exposure, leading to safer and more effective well interventions. Proper handling and material selection for CT bottom-hole assembly (BHA) components are vital for mitigating operational risks in highly sour and acidic settings. Following best practices and utilizing advanced materials where appropriate improves the durability and reliability of intervention operations.

Real-time inspection during active operations—enabled by multi-sensor integration and sophisticated data analytics—revolutionizes risk management, operational efficiency, and CT lifespan. The capability to monitor pipe condition in real-time offers unmatched safety, decreases non-productive time, and generates significant cost efficiencies by preventing failures and optimizing asset use.

References

1. Agha, K. R., Al-Shehri, D., & Al-Bagoury, M. (2012). *Qualification of 25Cr Super Duplex Coiled Tubing for Sour Service in High H2SEnvironment.* SPE Paper 156992. https://doi.org/10.2118/156992-MS
2. Al-Arnaout, I., Al-Hajji, H., Al-Fadhli, J., … & Al-Mumen, A. (2019). *Real-Time Monitoring and Control Enhance Coiled Tubing Acid Stimulation in Complex Sour Carbonate Reservoir.* SPE Paper 194987. https://doi.org/10.2118/194987-MS
3. Al-Mutairi, S. H., Hill, A. D., & Nasr-El-Din, H. A. (2008). Effect of H2S on the Rheological Properties of Emulsified Acids. *SPE Journal*, **13**(04), 413–421. https://doi.org/10.2118/102715-PA
4. API RP 5C7: 2017. *Recommended Practice for Coiled Tubing Operations in Oil and Gas Well Services* (3rd ed.). American Petroleum Institute.
5. Bello, O., Reinicke, K. M., & Teodoriu, C. (2014). State of the Art and Future Directions of Coiled Tubing Telemetry Systems. *SPE Production & Operations*, **29**(04), 285–294. https://doi.org/10.2118/163356-PA
6. Craig, S. (2003). *Development and Application of High-Strength Coiled Tubing in Sour Service*. SPE Paper 81727. https://doi.org/10.2118/81727-MS
7. Craig, S., & Brown, L. (2018). Coiled Tubing in Acid Stimulation. In *Coiled Tubing Operations Handbook* (pp. 237–265). Gulf Professional Publishing. https://doi.org/10.1016/B978-0-12-812933-8.00007-0
8. Elboujdaini, M., & Revie, R. W. (2009). Sulfide Stress Cracking. In *Uhlig's Corrosion Handbook* (3rd ed., pp. 221–230). John Wiley & Sons.
9. Fredd, C. N., & Fogler, H. S. (1998). The Influence of Chelating Agents on the Kinetics of Calcite Dissolution. *Journal of Colloid and Interface Science*, **204**(1), 187–197. https://doi.org/10.1006/jcis.1998.5554
10. Frenier, W. W. (2001). *Technology for Chemical Cleaning of Industrial Equipment*. NACE International.
11. Frenier, W. W. (2004). Development of Environmentally Friendly Corrosion Inhibitors for the Oil and Gas Industry. *Green Chemistry*, **6**(8), 369–374. https://doi.org/10.1039/B406149K
12. Frenier, W. W., Growcock, F. B., & Lopp, V. R. (2000). Mechanisms of Corrosion Inhibitors Used in Acidizing Wells. *SPE Production & Facilities*, **15**(03), 168–175. https://doi.org/10.2118/65384-PA
13. Heuer, J. K., & Stubbins, J. F. (1999). An XPS Characterization of FeCO3 Films from CO2 Corrosion. *Corrosion Science*, **41**(7), 1231–1243. https://doi.org/10.1016/S0010-938X(98)00180-2
14. ISO 15156-2:2015. *Petroleum and natural gas industries — Materials for use in H2S-containing environments in oil and gas production — Part 2: Cracking-resistant carbon and low-alloy steels, and the use of cast irons*. International Organization for Standardization.
15. Jovancicevic, V., Ramachandran, S., & Prince, P. (1999). Inhibition of Carbon Dioxide Corrosion of Mild Steel by Imidazolines and Their Precursors. *Corrosion*, **55**(5), 449–455. https://doi.Org/10.5006/1.3283988
16. Kelland, M. A. (2009). *Production Chemicals for the Oil and Gas Industry*. CRC Press.

17. Kermani, M. B., & Harrop, D. (1996). The Impact of Corrosion on the Oil and Gas Industry. *SPE Production & Facilities*, **11**(03), 186–190. https://doi.org/10.2118/29784-PA
18. Kermani, M. B., & Morshed, A. (2003). Carbon Dioxide Corrosion in Oil and Gas Production—A Compendium. *Corrosion*, **59**(8), 659–683. https://doi.org/10.5006/L3277596
19. Mahmoud, M. A., Nasr-El-Din, H. A., DeWolf, C. A., & LePage, J. N. (2011). Evaluation of a New Environmentally Friendly Chelating Agent for High-Temperature Applications. *SPE Journal*, **16**(03), 559–574. https://doi.org/10.2118/127923-PA
20. NACE MR0175/ISO 15156 (Multiple Parts). Petroleum and natural gas industries — Materials for use in H2S-containing environments in oil and gas production. NACE International / ISO.
21. NACE TM0177: 2016. Laboratory Testing of Metals for Resistance to Sulfide Stress Cracking and Stress Corrosion Cracking in H2S Environments. NACE International.
22. NACE TM0284: 2016. *Evaluation of Pipeline and Pressure Vessel Steels for Resistance to Hydrogen-Induced Cracking*. NACE International.
23. Nasr-El-Din, H. A., & Samuel, M. (2007). Laboratory Evaluation of Corrosion Inhibitors for Application in Sour Wells. *SPE Production & Operations*, **22**(04), 526–535. https://doi.org/10.2118/100474-PA
24. Obot, I. B., Umoren, S. A., & Gasem, Z. M. (2015). Corrosion Inhibition of Carbon Steel in Sour Media by Novel Nanoparticles. *Journal of Adhesion Science and Technology*, **29**(13), 1321–1336. https://doi.org/10.1080/01694243.2015.1026859
25. Ouyang, L., & Kuru, E. (2014). Real-Time Monitoring of Acid Stimulation Treatments in Carbonate Reservoirs Using DTS and DAS. *SPE Journal*, **19**(06), 1155–1164. https://doi.org/10.2118/168205-PA
26. Patton, L. D. (1990). Coiled Tubing Operations. In *Petroleum Engineering Handbook (SPE, Ch. 8)*. Society of Petroleum Engineers.
27. PRCI (Pipeline Research Council International). (2005). Materials Optimization for Sour Service Coiled Tubing. PRCI Report L52048.
28. Stanley, R. K., & Gentili, N. (1986). Failure Analysis of Coiled Tubing. *SPE Production Engineering*, **1**(06), 461–468. https://doi.org/10.2118/14122-PA
29. Tiratsoo, J. N. H. (2010). Coiled Tubing: Advanced Well Intervention and Drilling Technology. PennWell Books.
30. Tipton, S. M. (1994). Fatigue Life and Plastic Strain Behavior of Coiled Tubing. *SPE Drilling & Completion*, **9**(01), 15–20. https://doi.org/10.2118/23102-PA
31. OSHA 29 CFR 1910.120 - Hazardous Waste Operations and Emergency Response.
32. ANSI/ASSP Z390.1 - Accepted Practices for Hydrogen Sulfide Safety Training Programs.
33. NACE MR0175 / ISO 15156: Petroleum and natural gas industries — Materials for use in H_2S-containing environments in oil and gas production.
34. API RP 5C7: Coiled Tubing Operations in Oil and Gas Well Services.
35. API Standard 53: Blowout Prevention Equipment Systems for Drilling Wells.
36. API RP 14C: Analysis, Design, Installation, and Testing of Safety Systems for Offshore Production Facilities. (Principles applicable).
37. OSHA Standards (29 CFR 1910.1200, 1910.1000 Table Z-2, etc.) - Hazard Communication, Air Contaminants (H_2S)